



**Department of Information Technology**  
**Academic Year 2023 - 2024 (Odd Semester)**

**Degree, Semester & Branch:** III Semester B.Tech. IT

**Course Code & Title:** CS3362 & Foundations of Data Science

**Name of the Faculty member (s):** Dr.V.Anusuya, ASCP/IT

**Innovative Practice Description**

- **Unit / Topic:** Unit V / **Geographic Data with Base map**

- **Course Outcome:** CO5

- **Topic Learning Outcome:** TLO11

- **Activity Chosen:** Learning by Teaching

- **Justification:**

In unit V, the students need to understand the Geographic Data with Base map. This activity helps the students to understand the voluntarism and leadership.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

- A topic Geographic Data with Base map was given to Ms. Siva Sekar one week before the presentation.
- He prepared the presentation and taken the class on 16.12.2023.
- The topic was delivered around 20 minutes with demonstration given using Google colab.
- The students discussed with an example using the different attributes and styles.
- The implementation of Learning by teaching activity as shown in figure1 and figure2.

- CO – PO / PSO mapping:

| CO  | PO1 | PO2 | PO3 | PO4 | PO12 | PSO1 | PSO2 | PSO3 |
|-----|-----|-----|-----|-----|------|------|------|------|
| CO5 | 3   | 3   | 2   | 2   | 2    | 2    | 3    | 2    |

(1 – Low      2 – Moderate      3 – High)

- PO / PSO mapped:

| Innovative practice                  | PO1                                                                                      | PO2                                                                                         | PO3                                                                                   | PO4                                                         | PO12                                                                                           | PSO1                                                                                      | PSO2                                                                                                                                          | PSO3                                                                                                             |
|--------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
|                                      | 3                                                                                        | 3                                                                                           | 2                                                                                     | 2                                                           | 2                                                                                              | 2                                                                                         | 3                                                                                                                                             | 2                                                                                                                |
| <b>Justification for correlation</b> | Apply basic Knowledge and fundamentals of data science for Geographic data with base map | The students will be able to identify the need for analysis of complex engineering problems | Able to design solutions for any applications using visualization libraries in python | Able to analyze and apply visualization libraries in python | Ability to recognize the need for visualization libraries in python in real world applications | Exhibit knowledge in visualization libraries for data science for application development | Exhibit knowledge in visualization libraries for data science in the field of IoT, Cloud Computing and Data Analytics to provide IT solutions | Exhibit knowledge in visualization libraries for data science in software projects to deliver a quality product. |

- **Images / Screenshot of the practice:**



***Figure 1: Learning by Teaching Handled by Siva Sekar P***



***Figure 2: Learning by Teaching***

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Each student was listening to the class with keen attention. They understand the importance of data visualization.
- Students understand the geographic visualization using the basemap package.
- It makes the students to gain knowledge on different projections of a map.

- ❖ ***Benefit of the practice:***

- Students have the capacity of analyzing geographic data and identifying regions with rivers, coasts, and land.
- Students can able to visualize the geographic area with different colors and styles.

- This activity helps the students to know importance of visualization packages.

❖ ***Challenges faced in implementation:***

- Some students make noise during presentation initially, after demonstration with colab the students get interested.
- The topic was planned for 20 minutes, but he took 40 minutes.

**References:**

- ❖ <https://ohioline.osu.edu/>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/>

**Signature of Faculty Member**

**HOD**